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Assessing the Risk of Bitcoin Futures Market: New Evidence

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process and then consider this information in the HAR model. Both the in-sample and out-of-sample analyses show that jumps offer added information which is not provided by the existing HAR models. In addition, a novel finding is that the jump-induced volatility offers incremental information relative to the Bitcoin implied volatility index. In sum, our results indicate that the HAR-RV process comprising the leverage effects and jump volatility would predict the RV more precisely compared to the standard HAR-type models. These findings have important implications to cryptocurrency investors.

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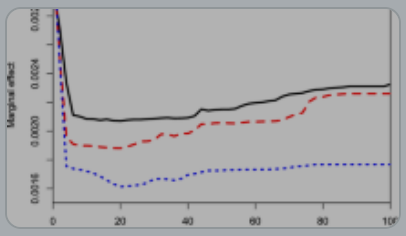

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1 Introduction

Cryptocurrencies represent decentralized payment systems involving technological innovation. Notably, such payment process provides cryptocurrencies an incomparable transparency. Transactions of digital currencies are all recorded on the open public ledger known as 'blockchain'. Note that in blockchain technology, data science guarantees that all the transactions are safe and protected. It ensures the integrity and security of blockchain transactions. In particular, data science effectively detects any sort of suspicious activities on the blockchain network. Besides, it also assures that transactions are executed promptly. Due to this transparency and 24-h accessibility, digital currencies have gained immense popularity among the investors [1].

Over the past few years, digital currencies have grown rapidly in price, attractiveness, and mainstream adoption [2]. As of July, 2021, there are over 6000 cryptocurrencies being traded and currently, the global market capitalization amounts to $1.6 T. Among these digital currencies, Bitcoin still leads the market with a total market capitalization of 700bn, corresponding to 44% of the overall market capitalization.

Given the popularity and market capitalization of this leading crypto, it is of paramount importance to precisely predict its volatility so that investors who take large positions in it

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Therefore, the patterns of volatility for this market have received ample attention among the academics, investors and policymakers [9].

The earlier works assessing the volatility dynamics of BTCF can be divided into three different categories. The first group of literature is mainly focused on how the volatility of BTCF impacts the price and volatility of other cryptocurrencies [10, 11]. The second category examines the quantitative risk management of BTCF market [12, 13], while the last category has assessed its hedging effectiveness [14, 15].

The present study joins the existing literature to propose an extended heterogeneous autoregressive (HAR) process which could be beneficial for predicting the realized volatility (henceforth, RV) of the BTCF market more precisely. It is worth mentioning that while a growing body of literature has explored the volatility dynamics and risk management of the BTCF market, prediction of realized volatility for this asset remains understudied. Given that volatility plays a key role in making appropriate asset allocation decisions and managing the risk of investor portfolio, finding accurate estimates of future RV for the BTCF market is crucial. To this end, this paper aims to fill this void in the existing literature by investigating the role of time-varying jumps occurring in BTCF returns when predicting the RV using the HAR-type models.

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properly. Hence, our analysis could be useful for deriving appropriate asset pricing models which could minimize the risk linked to cryptocurrency investments.

Second, we check the predictive ability of Bitcoin implied volatility index (henceforth, BVIN) when modeling the RV of BTCF market. It is now well-documented in the finance literature that the implied volatility offers incremental information for equity market volatility forecasts [18]. However, such literature is scarce in the context of cryptocurrency markets. We thus extend the prior literature by investigating whether the information content of BVIN is useful for the volatility forecasts of the BTCF market. In doing so, we also verify if the HAR process including the volatility of time-varying jumps outperforms the HAR model containing the information on the implied volatility index.

In our empirical analysis, we obtain the estimates of (time-varying) jump volatility by applying the GARCH-jump process to BTCF returns. We then use the information content of jump volatility in the HAR model to forecast the RV of BTCF returns. Our findings indicate the importance of jump-induced volatility. In fact, the HAR approach considering the information on jump-induced volatility outperforms all other models used in this study. Both in-sample and out-of-sample analyses support this novel finding. Our analysis thus reveals that financial time series alone would include predictive contents for forecasting the future RV and that using the jump process could be useful in this regard. Given that precise

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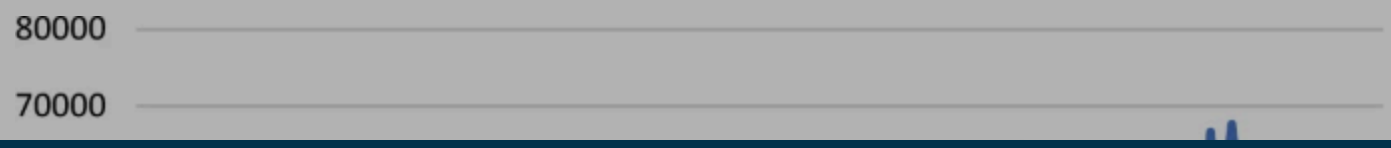
Table 1 shows the descriptive statistics for the Bitcoin futures index. The results suggest that returns are negatively skewed with a leptokurtic distribution. The Jarque–Bera (J-B) test also shows the violation of normality assumption. Finally, the augmented Dickey Fuller (ADF) and Philips Perron (PP) tests confirm that Bitcoin futures returns appear to be stationarity.

Table 1 Summary statistics

Figure 1 exhibits the bitcoin futures prices for the sample period considered in this paper. It demonstrates that although the prices seem to be low initially, the market experiences substantial increments in 2021. However, we also witness a notable drop after the inception of Russo-Ukrainian war.

Fig. 1



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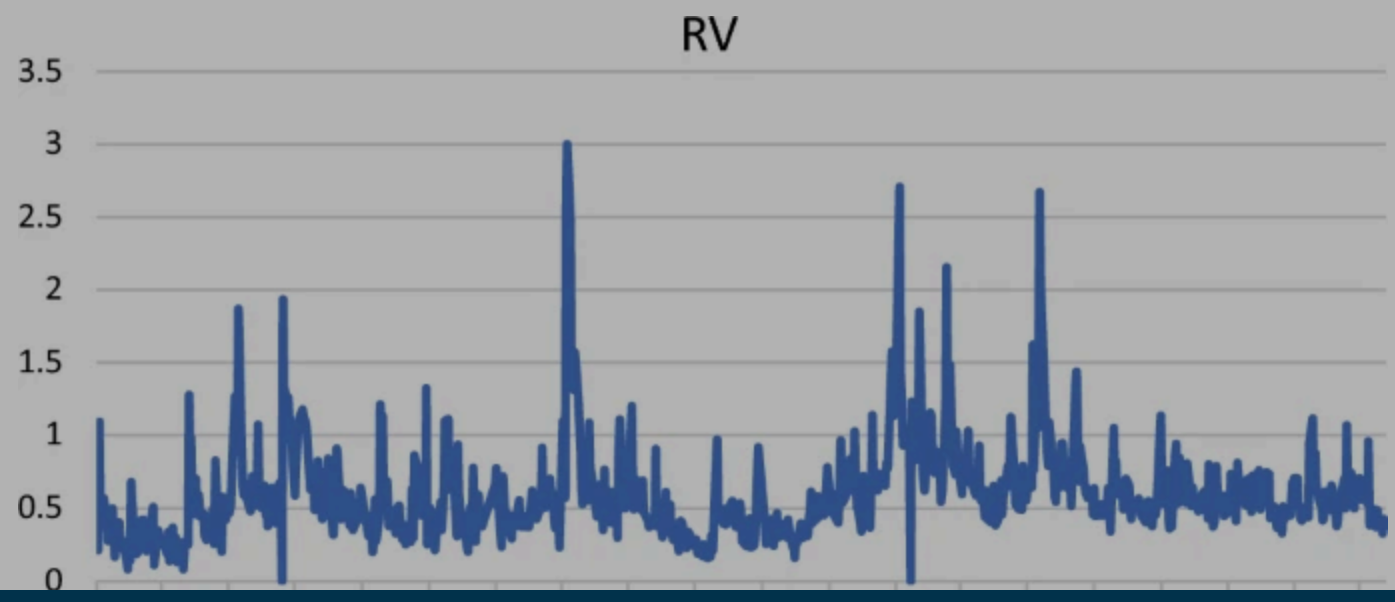
Time series plot of Bitcoin futures prices

Next, Fig. 2 plots the realized volatility of Bitcoin futures prices. This graph reveals that while the futures index is, in general, volatile, the risk increases substantially during the COVID-19 pandemic periods.

Fig. 2



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3.1 GARCH-Jump Approach

Following Chan and Maheu [16], we employ the GARCH-jump model as follwos[3]:

$$ r_{t} = \pi + \mu r_{t - 1} + \epsilon_{t} $$

(1)

where \({r}_{t}\) indicates the logarithmic difference for the BTCF index at time *t*, and \({\epsilon }_{t}\) refers to the innovation term which is specified as:

$${ \epsilon }_{t}={\epsilon }_{1t}+{\epsilon }_{2t}$$

(2)

where \({\epsilon }_{1t}\) will follow the GARCH (1,1) specification:

$$ \smallint_{1t} = \sqrt {h_{t} } z_{t} , z_{t} \sim NID\left({0,1} \right) h_{t} = \omega + \alpha \epsilon_{1t - 1}^{2} + \beta h_{t - 1} $$

(3)

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(5)

With an autoregressive conditional jump intensity (ARJI) given as[4]:

$${\lambda }_{t}={\lambda }_{0}+\rho {\lambda }_{t-1}+\gamma {\xi }_{t-1}$$

(6)

In Eq. 6, \({\lambda }_{t}\) indicates the time-varying conditional jump intensity parameter, \({\lambda }_{0}\) is the constant jump intensity, and \({\xi }_{t-1}\) denotes the intensity residual. Chan and Maheu [16] assume that \({\lambda }_{t}>0\), \({\lambda }_{0}>0\), \(\rho >0,\) and \(\upgamma >0\).

We define the log-likelihood as:

$$L\left(\Theta \right)=\sum_{t=1}^{T}{\text{log}}f(\left.{X}_{t}\right|{I}_{t-1};\Theta)$$

where \(\Theta =(\uppi ,\mu ,\upomega ,\mathrm{\alpha },\upbeta ,\uptheta ,\vartheta ,{\lambda }_{0},\uprho ,\upgamma)\) and \({I}_{t-1}\) is the information set.

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[23]. Previous studies [24,25,26,27] also demonstrate its dominance over the GARCH-type, SV-type, VAR-RV, MIDAS-RV, and ARFIMA-RV models.

Following Corsi [28] and Busch et al. [29], we define the baseline HAR-RV model as follows:

3.2.1 HAR-RV

$${RV}_{t,t+h}={\tau }_{0}+{\tau }_{d}{RV}_{t}+{\tau }_{w}{RV}_{t-5,t}+{\tau }_{m}{RV}_{t-22,t}+{\varepsilon }_{t}$$

(8)

with *h* being equal to 1, 5 and 22 depending on the daily, weekly and monthly volatility components, respectively and

$${RV}_{{t}_{1},{t}_{2}}=\frac{1}{{t}_{2}-{t}_{1}}\sum_{t={t}_{1}+1}^{{t}_{2}}{RV}_{t}$$

(9)

In our paper, a number of extensions to the baseline HAR-RV model have been considered.[5]

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({r}_{t-22,t}^{-}={\text{min}}\left(({r}_{t-21}+{r}_{t-20}+\dots + {r}_{t}\right)/\mathrm{22,0})\).

As mentioned earlier, the HAR model is also extended using the information content of BVIN index. This process, called LHAR-RV-IV, is given as:

3.2.3 LHAR-RV-IV

$${RV}_{t,t+h}={\tau }_{0}+{\tau }_{d}{RV}_{t}+{\tau }_{w}{RV}_{t-5,t}+{\tau }_{m}{RV}_{t-22,t}+{\psi }_{d}{r}_{t}^{-}+{\psi }_{w}{r}_{t-5,t}^{-}+{\psi }_{m}{r}_{t-22,t}^{-}+\delta {BVIN}_{t}+{\varepsilon }_{t}$$

(11)

Next, we introduce the jump-induced volatility (JV) term, specified in Eq. (7), to the LHAR-RV process as follows:

3.2.4 LHAR-RV-JV

$${RV}_{t,t+h}={\tau }_{0}+{\tau }_{d}{RV}_{t}+{\tau }_{w}{RV}_{t-5,t}+{\tau }_{m}{RV}_{t-22,t}+{\psi }_{d}{r}_{t}^{-}+{\psi }_{w}{r}_{t-5,t}^{-}+{\psi }_{m}{r}_{t-22,t}^{-}+{\phi }_{d}{JV}_{t}+{{\phi }_{w}{JV}_{t-5,t}+{\phi }_{m}{JV}_{t-

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Notably, our objective is to forecast one-month ahead volatility only given that options expire at a monthly frequency. Hence, h equals 22 in our analysis. Busch et al. [29] also advocate this when forecasting the RV of bond, currency and equity markets using the VIX index.

3.3 Out-of-Sample Forecast s

3.3.1 Forecast Evaluation

The forecasting performance of different models used in this study is evaluated using the heteroskedasticity adjusted root mean square error (HRMSE) proposed by Bollerslev and Ghysels [33]. We define this statistic as:

$$HRMSE=\sqrt{\frac{1}{T}\sum_{t=1}^{T}{\left(\frac{{RV}_{t}-\widehat{{RV}_{t}}}{{RV}_{t}}\right)}^{2}}$$

(14)

With T indicating the number of data points to be forecasted, while ${RV}_{t}$ and $\widehat{{RV}_{t}}$ specify the true and estimated volatility for day t, respectively.

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$\varvec{f}\left(\varvec{e}_{1\varvec{t}}\right)-\varvec{f}(\varvec{e}_{2\varvec{t}})$, with $\varvec{f}\left(.\right)$ denoting a function of forecast errors. We then wish to test:

$$H_{0}:E\left(d_{t}\right)=0$$

DM [34] shows that the approximate asymptotic variance of $\overline{d }$ is given as:

$$Var\left(\overline{d }\right)\approx k^{-1}\left[\eta _{0}+2\sum_{l=1}^{p-1}\eta _{l}\right]$$

(16)

where $\eta _{l}$ indicates the l-th autocovariance of d_{t}, which is estimated as:

$$\widehat{\eta }=k^{-1}\sum_{t=l+1}^{k}(d_{t}-\overline{d })(d_{t-l}-\overline{d })$$

(17)

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provide incremental information relative to the baseline HAR-RV model. We define the MZ regression approach as follows:

$${RV}_{t}={\varphi }_{0}+{\varphi }_{1}\widehat{{RV}_{t}}+{\epsilon }_{t}$$

(19)

where, \({RV}_{t}\) and \(\widehat{{RV}_{t}}\) are the true and estimated volatility for day *t*, respectively. We then compare the HAR models on the basis of R^2 (coefficient of determination) statistics. Notably, our in-sample estimation period spans from January 2019 to March 2021 and the out-of-sample period from April 2021 to March 2022.

4 Empirical Findings

4.1 In-Sample Estimates

The findings of the HAR models are shown in Table 3. Before discussing these results, we briefly explain the estimates of out GARCH-jump approach presented in Table 2. It is evident from these estimates that jumps exist in BTCF returns as the parameters of the ARJI process are mostly significant. Notably, the positive coefficient of the jump variance (see the

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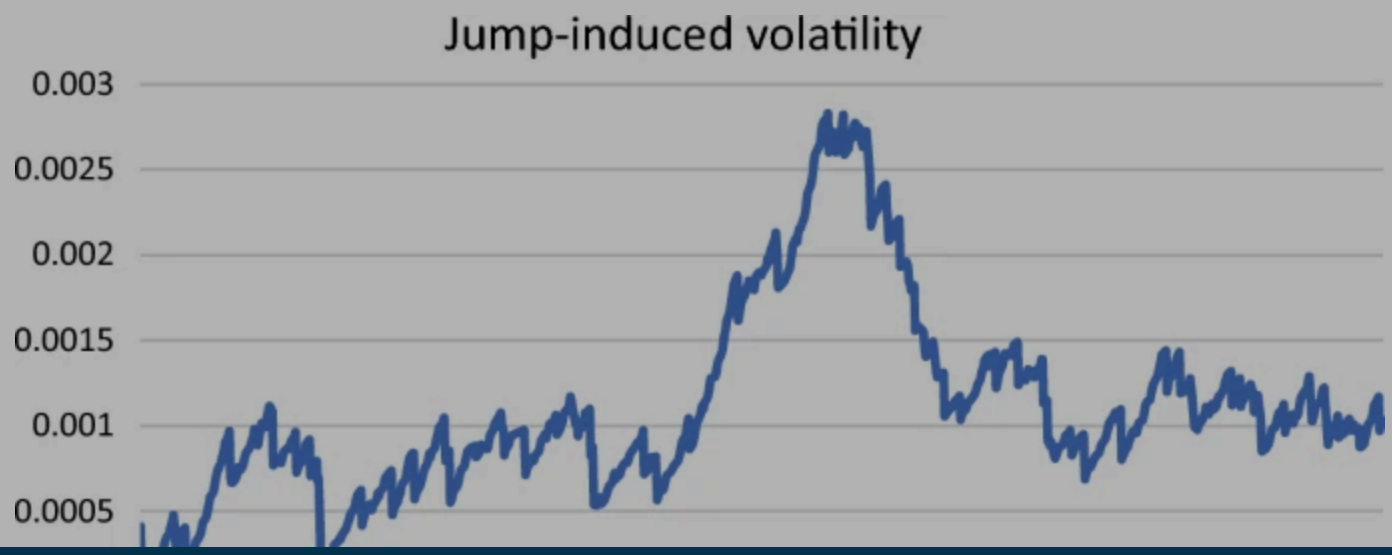
Table 2 Estimates of the GARCH-ARJI model

Moreover, Fig. 3 depicting the jump-induced volatility also confirms presence of high jumps during the COVID-19 crisis periods. Hence, time-varying jumps may include predictive content for forecasting the RV of BTCF market.

Fig. 3



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not increase the $\mathbf{R}^{2}$ statistic substantially. This is not consistent with earlier studies given that implied volatilities, in general, contain predictive information for the financial markets. Forth, inserting the JV factor to the LHAR-RV model increases the $\mathbf{R}^{2}$ statistic markedly, revealing the importance of using the information on jumps when modeling the RV of BTCF market. Fifth, the RV is sensitive only to the monthly component of the jump-induced risk factor revealing the long memory behavior of volatility. Summarizing, our analysis suggests that including leverage effects and time-dependent jumps in the HAR-RV model would increase its accuracy.

Table 3 Estimates of different HAR-RV models

4.2 Out-of-Sample Analysis

The out-of-sample forecast results based on the HMSE and MAE statistics along with MZ regression model are exhibited in Table 4. Our in-sample estimation period spans from January 2019 to March 2021 and the out-of-sample period from April 2021 to March 2022. The results reveal that the LHAR-RV-IV-JV process produces the lowest HRMSE and MAE statistics. The Diebold and Mariano (DM) [34] test further supports these findings by

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LHAR-RV-IV-JV) surpass other approaches by yielding higher $\({{\text{R}}}^{2}\)$ values. Our overall results thus suggest that the information on time-dependent jumps in BTCF returns is essential for increasing the precision of the HAR-RV model. Therefore, participants in Bitcoin futures market should analyze such jumps while forecasting the realized volatility of this new asset class.

4.3 Forecasting Value-at-Risk (VaR)

We now conduct ta simple VaR analysis for finding the best forecast model. In doing so, we test for all the HAR-RV models with a VaR estimated for the quantile level $\(q\)$. To this end, the likelihood ratio (LR) test, developed by Kupiec [39], is applied.

We begin with the following hit sequence:

$${Hit}_{t}=\left\{\begin{array}{c}1, \text{ if } {r}_{t}<{VaR}_{t}\\ 0, \text{ if } {r}_{t}\ge {VaR}_{t}\end{array}\right.$$

(20)

Where $\({r}_{t}\)$ refers to the return on day t and $\({VaR}_{t}\)$ is specified as:

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$$LR=-2{\text{ln}}\{{\left(1-q\right)}^{N}{q}^{T-N}/{\left(1-N/T\right)}^{T-N}{\left(N/T\right)}^{N}\}\sim {\chi }^{2}(1)$$

(22)

The p-values of this test are provided in Table 5. Our findings show failure rates for both left and right quantiles. Given that the accuracy of the HAR models increases with the increment in p-values, these numbers further reveal the significance of jumps in forecasting the RV of BTCF market. Overall, our analysis suggests that both leverage effects and time-varying jumps are crucial for predicting the VaR with precision.

Table 5 Forecasting value-at-risk

5 Conclusions

This paper aims to forecast the realized volatility of Bitcoin futures market. In doing so, we propose an augmented HAR-RV model to consider the information on time-dependent jumps observed in BTCF returns. In particular, we estimate the jump-induced volatility

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difficult to fully understand the role of such jumps for predicting the risk of cryptocurrency investments. As investors and policymakers are very keen to have further knowledge about this domain, future research could employ more sophisticated statistical tools to shed more light on how the information on jump-induced volatility can be used to obtain improved volatility forecasts during the crisis periods. To this end, applications of big data methods and artificial intelligence such as neural networks could provide superior forecasts for Bitcoin volatility. Artificial neural networks are particularly useful for handling high frequency trading data and predicting financial risk. Given that real-time decision making also benefits from the use of artificial intelligence deep learning methods are highly recommended for forecasting the price of Bitcoin in real-time. Prior studies such as Shi [43], Olson and Shi [44], Shi et al. [45] and Tien [46] discuss a number of important big data methods which are quite useful for this sort of analysis. Note that future studies could also investigate whether jumps in Bitcoin returns can predict the volatility of other cryptos. In addition, attempts could be made to identify the key determinants of jumps occurring in Bitcoin price index. Given that jump-induced volatility represents a large fraction of market volatility, identifying such determinants might be essential for hedging cryptocurrency market risk amid the crisis periods.

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predicting future price movements based on prior data. Such forecasts are important for cryptocurrency investors when estimating volatility and formulating effective trading strategies. In sum, data science has emerged as a key tool for interpreting complex market behaviors, predicting trends and managing portfolio risk.

2. Jumps are often observed in Bitcoin market. Previous studies have also documented the presence of time-varying jumps in cryptocurrency markets [17, 19, 20]. In this paper, unlike the prior works, we study the occurrence of such jumps in Bitcoin futures market and their role in forecasting the RV.

3. The choice of the AR(1) specification is based on the AIC and BIC values.

4. The ARJI process is an extension of the constant jump intensity (CJI) process, which is proposed by Jorion [21]. In the CJI process, it is assumed that $\lambda_{t}=\lambda_{0}$, implying that jump intensity is independent of time. However, a number of recent studies [22, 23] document that such intensity is time-dependent and hence prefer ARJI model to CJI process.

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Ethics declarations

Ethical Statement

This work was not copied from any source, and it does not harm humans or society in any way.

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Keywords

Bitcoin futures market | Realized volatility | Jump-induced volatility | Bitcoin implied volatility index | Leverage effects | HAR-RV models

JEL Classification

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